MATH9102

LINEAR REGRESSION

Sources used in creation of this lecture:
Statistics and Data Analysis, Peck, Olsen and Devore; Discovering Statistics Using R, Field, Miles and Field; Understanding Basic Statistics, Brase and Brase;SPSS Survival Manual, Julie Pallant

Predictive Models

Used to explore the relationship between one outcome variable and a set of independent variables (predictors)

You should have a sound theoretical or conceptual reason for exploring the relationship and the order of the variables entering the model

For variables that have not been previously shown to contribute to the variation in the outcome variable try to establish evidence (statistical or research) for their inclusion in advance

Predictive Modelling - Linear

Linear models describe relationships between variables that are correlated in a linear way

- When one variable changes, the other tends to change proportionally (either positively or negatively).

The model assumes an outcome variable (response/dependent) that you want to predict, and one or more independent variables (predictors) that help explain it.

Predictive Modelling - Linear

It applies a best-fit line (or plane, if multiple predictors) through the data to minimize the distance between the observed values and the line — this is usually done by ordinary least squares (OLS).

Once fitted, the model can be used to predict future values of the dependent variable given new values of the independent variables.

Predictive Modelling - Linear

Equation form:

$y=\beta 0 + \beta 1x1 + \beta 2x2 + \cdots + \beta nxn + \varepsilon$

Where:

- y is the output variable
- xi are independent variables
- βi are coefficients
- and ε is the error term.

Predictive Modelling - Linear

Linear models don't just describe relationships — they can also be used to **quantify differences** between groups or conditions by estimating how much the dependent variable changes when one factor changes.

You incorporate the **investigation of difference** by including **categorical variables** (e.g. gender, region, treatment type) as **factors** in the model.

Predictive Modelling - Linear

Conceptually:

- Prediction focuses on estimating future outcomes.
- Investigation of difference focuses on comparing predictions for groups or conditions
  - e.g. “Is the prediction for Group A significantly different from Group B?”
- Both use the same linear model structure, but the interpretation of coefficients changes:
  - Continuous predictors → rate of change
  - Categorical predictors → group differences

Regression Model

Explanatory (independent/predictors) variables are used to model the behaviour of a response (outcome/dependent) variable

Most commonly used class of models in the statistical toolbox

Different types of regression suit different types of application

Predictive model – what does it allow you do?

Prediction

- Really what we are looking at is the variance in the outcome variable and how much of the variance could be considered to be explained by the predictor variables

Questions it allows you to answer:

- How well a set of variables can predict an outcome variable
- Which variable in a set is the best predictor
- Whether a variable is still able to predict an outcome when controlling for other variables

Predictive model – what does it allow you do?

The same model structure can be used to explore **differences between groups or conditions** — this is often called **testing for group effects**.

Here, we use categorical predictors (e.g. gender, region, treatment type) to ask whether belonging to one group leads to a statistically significant difference in the outcome variable compared to another group.

Questions it allows you to answer:

Do different groups (e.g. treatment vs. control) show significant differences in the outcome variable?

How large is the difference between these groups, after accounting for other predictors?

Before Regression

We need to establish evidence to support going ahead with building a predictive model

If we are asserting a relationship

We need to investigate if there is any evidence of a relationship using correlation and make a decision based on the results (strength, direction etc.)	Correlation – Pearson, Spearman, Kendall, Chi, Fisher

If we are asserting a differential effect for different groups

We need to investigate if there is any difference using either the appropriate test and make a decision based on the result	Difference – T-test (independent or paired), Mann-Whitney, Anova (one-way or multiple), Kruskal-Wallis, Friedmann, McNemar, Chi, Fisher

Causal Relationship

To assert a causal relationship is to claim that changes in the independent variable(s) create changes in the dependent variable

In practice can only assert that one factor causes change in another when you can satisfy the following criteria:

Association	**Time order**	**Non-spuriousness**

Causal Relationship - Association


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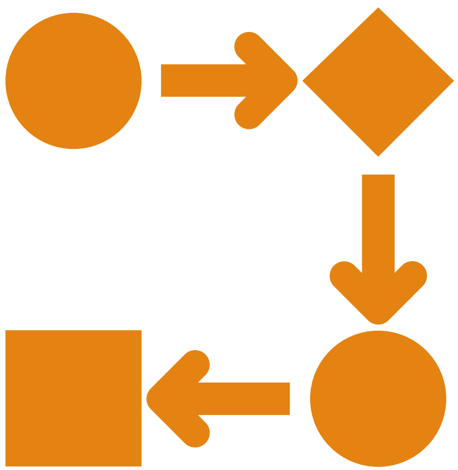
Variables correspond to each other in predictable ways

- Can be either positive (an increase in one causes an increase in the other)
- Or negative (an increase in one causes a decrease in the other)

Many associations are not causal

- E.g.
  - Red cars do not cause more accidents
  - Rather aggressive drivers are more likely to buy red cars
  - Aggressive drivers are more likely to be involved in accidents

Causal Relationship – Time Order



Logically the change in the independent variable *precedes* the change in the dependent

Need to be careful

- Even if time order is satisfied, a causal relationship may not exist

Causal Relationship – Non-spuriousness

A spurious relationship exists when two variables appear to be causally related but are not

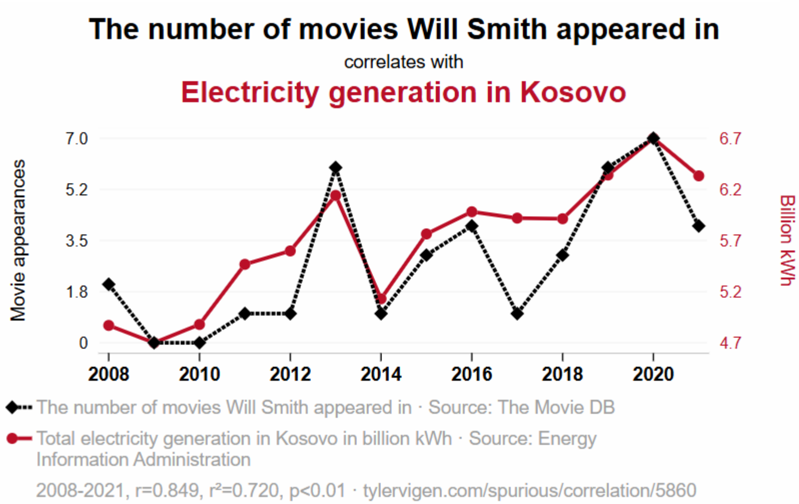
- Any observed dependencies between variables are due to chance or are both related to some unseen confounding factor

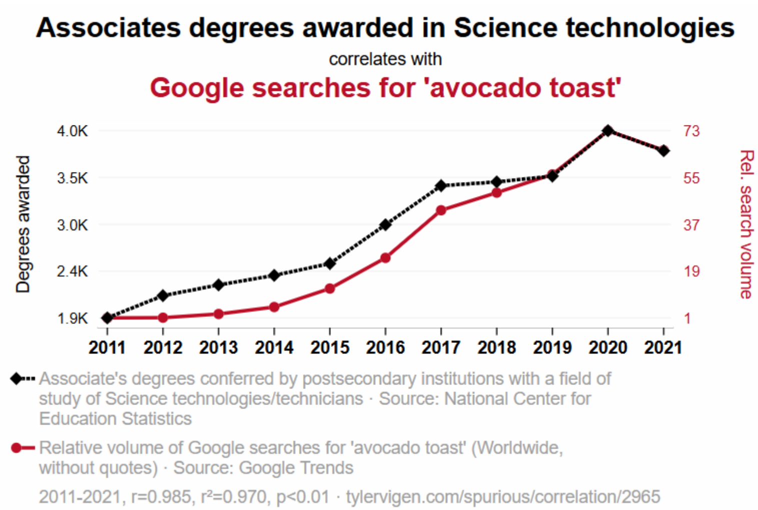
E.g.

- Ice cream consumption causing drowning deaths
- There is an association between ice cream consumption and drowning (more people die during times when a lot of ice cream is being consumed)
- Time order can also be satisfied – increase in ice cream sales precede increase in drowning deaths
- However, the unmeasured factor here is seasonal temperature
  - In warm weather more people consume ice cream and swim

Spurious Correlations

https://www.tylervigen.com/spurious-correlations





Relationships and Causality

Causality

- Determining how particular sets of conditions (represented by variables) lead to outcomes (also represented by variables)

Seldom are relationships **deterministic**

- Everyone completing a university degree will have a higher income than those who do not (unlikely to be true)

Most are **probabilistic**

- i.e. factors increase or decrease tendency towards particular outcomes
- Those with a university degree *tend* to earn more than those who do not
- University education is a *factor* that pushes another factor (income) in a predictable direction

Correlation and Causality

The third-variable problem:

- Causality between two variables cannot be assumed because there may be other measured or unmeasured variables affecting the results.

In our example

- Perceived stress does relate significantly to perception of control.
- There may be other factors that are influencing perception of control.
  - Some of these variables will have been measured by the researcher.
    - Some may not

Correlation and Causality

Direction of causality:

- Correlation coefficients say nothing about which variable causes the other to change.
- The correlation coefficient doesn't indicate in which direction causality operates.
- So, although it is intuitively appealing to conclude that perceived stress causes perceived control to change, there is no *statistical* reason why perceived control cannot cause perceived stress.
- This is where your knowledge of the constructs and existing theories in your area come in to play
  - You interpret your statistical findings through the lens of your subject area
  - Why including relevant background and related research is important in your reporting

Partial Correlations

Measures the relationship between two variables, **controlling** for the effect that a third variable

- Usually, a variable you suspect is influencing your two variables of interest
  - This can artificially inflate the correlation co-efficient found

Partial Correlation

Using survey.dat

- Julie Pallant

Interested in exploring the relationship between scores on the Perceived Control of Internal States Scale (tpcoiss) and scores on the Perceived Stress Scale (tpstress) but controlling for what is known as *socially desirable responding bias* (tendency to present yourself in a socially desirable way)

- This tendency is measured using Marlowe-Crowne Social Desirability Scale (tmarlow)

Partial Correlation

```
#perception of control and stress controlling for social desirability
ppcor::spcor.test(ydata$tpcoiss, ydata$tpstress, ydata$tmarlow)
```

```
##     estimate      p.value statistic   n gp  Method
## 1 -0.5269534 1.862222e-31 -12.69154 422  1 pearson
```

```
#perception of control and optimism controlling for social desirabilit
ppcor::spcor.test(ydata$tpcoiss, ydata$toptim, ydata$tmarlow)
```

```
##    estimate      p.value statistic   n gp  Method
## 1 0.4830465 5.348979e-26  11.29257 422  1 pearson
```

```
#stress and optimism controlling for social desirability
ppcor::spcor.test(ydata$tpstress, ydata$toptim, ydata$tmarlow)
```

```
##     estimate     p.value statistic   n gp  Method
## 1 -0.4412444 1.73877e-21 -10.06483 422  1 pearson
```

Zero order correlations

```r
#Get zero order correlations as well to fully explore the effect
cor(ydata$tpcoiss, ydata$tpstress)
```

```
## [1] -0.5812413
```

```r
cor(ydata$tpcoiss, ydata$toptim)
```

```
## [1] 0.5161727
```

```r
cor(ydata$tpstress, ydata$toptim)
```

```
## [1] -0.4667559
```

Reporting Results

Partial correlation was used to explore the relationship between perceived control of internal states (as measured by PCOISS) and perceived stress (measured by the Perceived Stress Scale), while controlling for scores on the Marlowe-Crowne Social Desirability Scale. Preliminary analyses were performed to ensure no violation of the assumption of normality, linearity and homoscedasticity. There was a strong, negative partial correlation between perceived control of internal states and perceived stress, controlling for social desirability, ($r=-.53$, $n=422$, $p<.001$). An inspection of the zero-order correlation ($r=-.58$) suggested that controlling for social desirability had very little effect on the strength of the relationship between these two variables.

Building a Linear Regression Model

MATH9102-W8-REGRESSION.QMD

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Statistics and Data Analysis, Peck, Olsen and Devore; Discovering Statistics Using R, Field, Miles and Field; Understanding Basic Statistics, Brase and Brase;SPSS Survival Manual, Julie Pallant

Simple Example – Linear Regression

Suppose we want to look at what variables predict a child's maths score aged 16 in the UK (GCSE)

We have a theory that their achievement on a standard maths test at aged 7 is a good predictor

- (We've seen this before in correlation…)

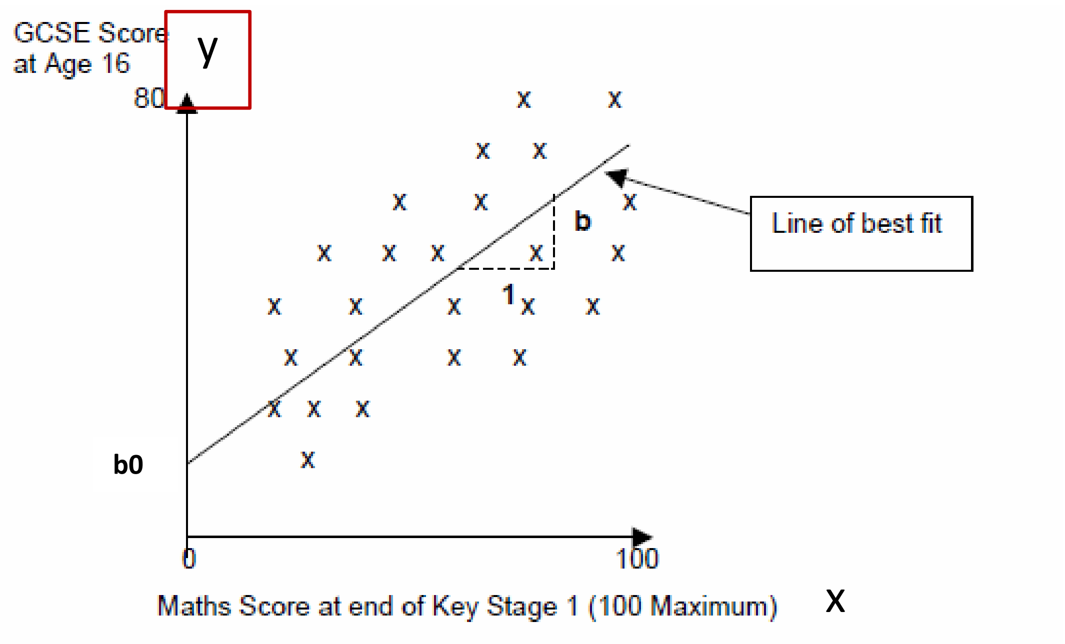
Simple Example

For a simple one predictor model.

'y' is the response variable to be predicted (in this case GCSE Score) – the dependent variable

This is a simple correlation.

$$y = b0 + bx + e$$

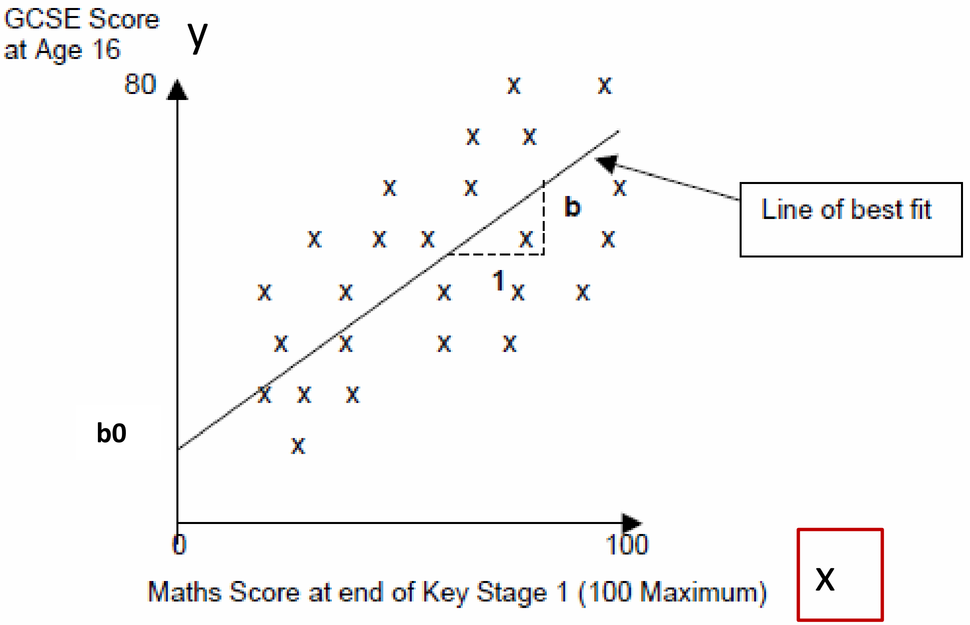


Relationship between Maths score age 7 with GCSE result at age 16 (for 25 students)

Simple Example

'x' is the predictor variable (in this case Maths Score aged 7) – independent variable

$$y = b0 + bx + e$$

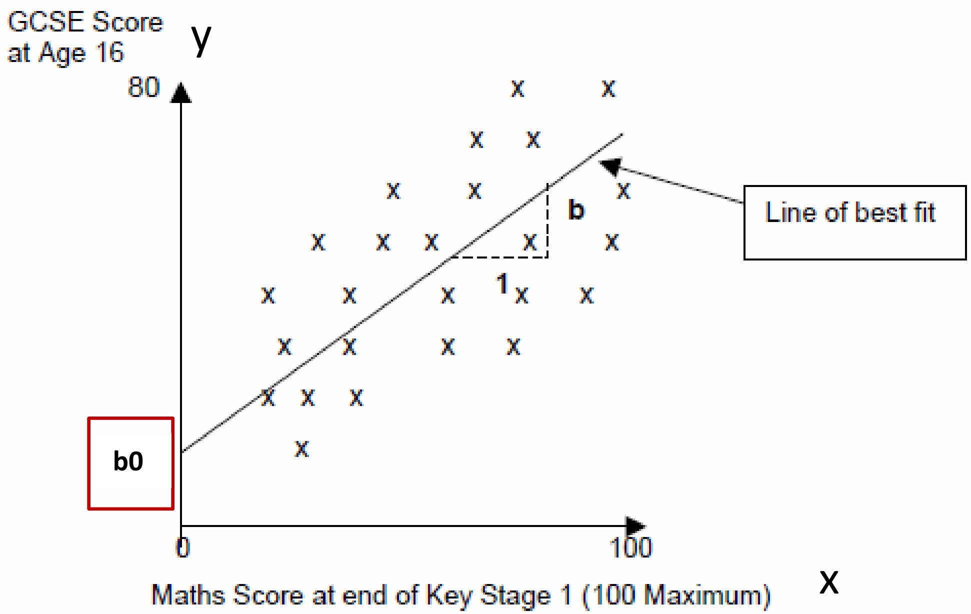


Relationship between Maths score age 7 with GCSE result at age 16

Simple Example

'b0' is the intercept or the point where the line crosses the y-axis;

$$y = b0 + bx + e$$



Relationship between Maths score age 7 with GCSE result at age 16

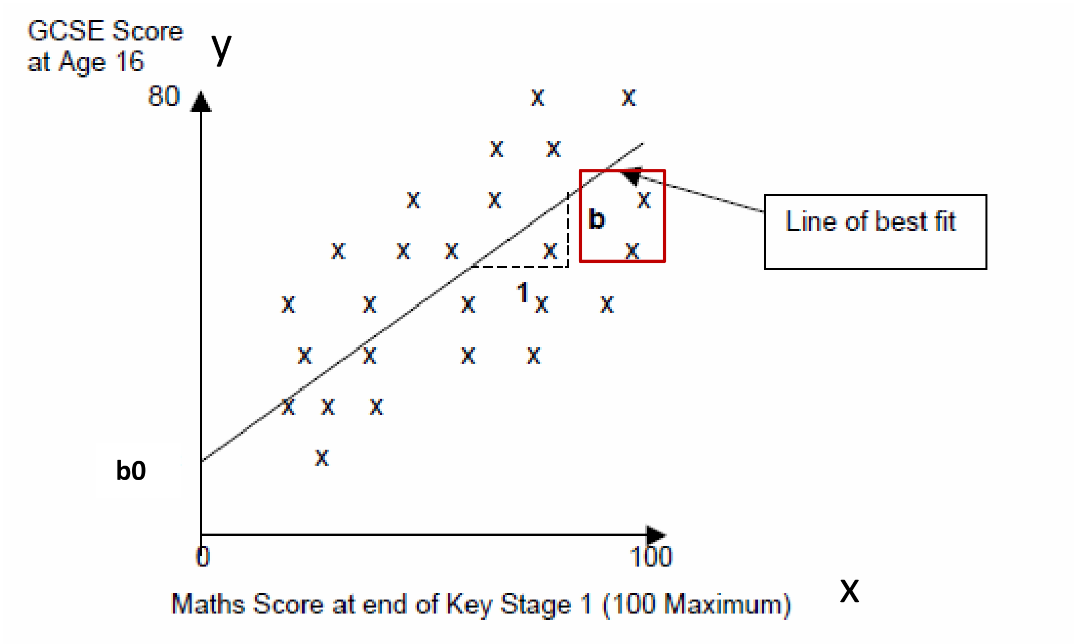
Simple Example

'b' is the gradient of the line

Represents the amount that the response variable changes for one unit change in the predictor variable

(e.g. for every one percentage point increase in a child's Maths Test score, the line suggests that the child's predicted GCSE Score will increase by 'b' points);

$$y = b0 + bx + e$$



Relationship between Maths score age 7 with GCSE result at age 16

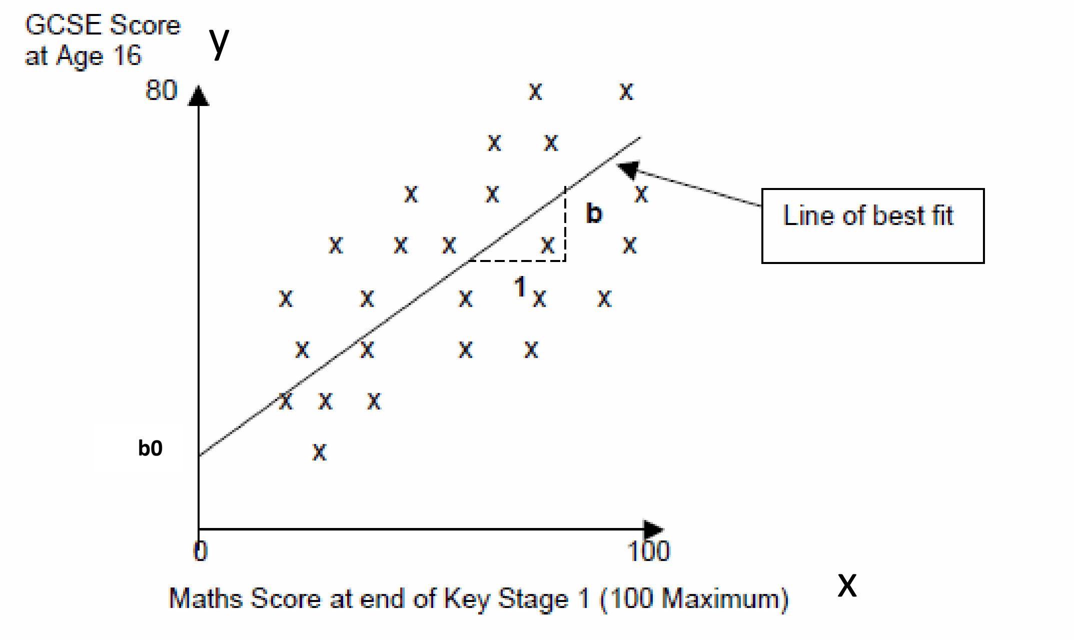
Simple Example

If the line did completely model the data then all of the points would rest exactly on the line.

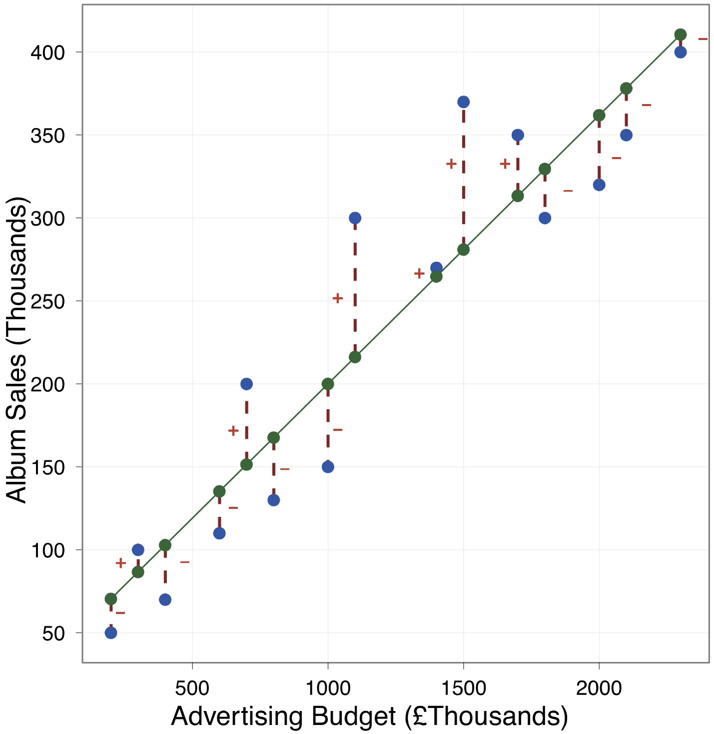
Because there is a difference between what the line predicts (expected values) each student will achieve in their GCSE Score and what they actually achieved (observed values).

This difference is basically the vertical distance between each point and the line itself and this distance is 'e' .

$$y = b0 + bx + e$$



Relationship between Maths score age 7 with GCSE result at age 16



A scatterplot of some data with a line representing the general trend. The vertical lines (dotted) represent the differences (or residuals) between the line and the actual data

Residuals

Using our model

We can re-write the overall model as an equation:

GCSE Score = b0 + (b x Maths Score) + error

To use the model to predict a child's future GCSE Score we would tend to drop the 'e' term and thus use the formula:

Predicted GCSE Score = b0 + (b x Maths Score)

Note: We are now referring to the response variable as the *'Predicted GCSE Score'* rather than the actual GCSE Score.

Using our model

Suppose we have plugged our numbers into our statistical tool, and it has given us the following values:

- b0 = 15 ; b = 0.7

Our model becomes:

- Predicted GCSE Score = 15 + 0.7 x Maths Score

Based on our data this represents our best estimate of what a child's future GCSE Score is going to be, *on average*, given their Maths Score at age 7.

- It is not going to be 100% correct as we know there is an error term.

Using our model

So suppose a child got just 10% in their maths test aged 7.

- Predicted GCSE Score = 15 + 0.7 x 10 (maths test aged 7)
- = 15 + 7 = 22.0

A child who gets 60% in their maths test aged 7:

- Predicted GCSE Score = 15 + 0.7 x 60
- = 15 + 42= 57

Looking at a student who gets 61% we can see what the gradient b is doing in practice:

- Predicted GCSE Score = 15 + 0.7 x 61
- = 15 + 42.7= 57.7

From our model we can see that on average one unit increase in Maths score aged 7 is predicted to lead to a 0.7 increase in GCSE Maths score aged 16.

Using our model

The gradient therefore represents the average increase in the response variable (GCSE Score) with one unit increase in the predictor variable (Maths Score).

Sample Dataset (Regression.sav)

Variable	Description
School	A unique numeric identifier for each school
Student	A unique numeric identifier for each student
Normexam	Student's exam score at age 16, normalised to have approximately a standard Normal distribution and a mean of 0 and standard deviation of 1. (Note that the normalisation was carried out on a larger sample, so the mean in this sub-sample is not exactly equal to 0 and the variance is not exactly equal to 1).
Cons	A column of 1's. This is used in the multilevel modelling package MLwiN to represent the intercept in a statistical model.
Standlrt	Student's score at age 11 on the London Reading Test (LRT), standardised using Z-scores.
Girl	1 = girl, 0 = boy
Schgend	School's gender (1 = mixed school, 2 = boys' school, 3 = girls' school)
Avslrt	Average LRT score in school
Schav	Average LRT score in school, coded into 3 categories (1 = bottom 25%, 2 = middle 50%, 3 = top 25%)
Vrband	Student's score in test of verbal reasoning at age 11, coded into 3 categories (1 = top 25%, 2 = middle 50%, 3 = bottom 25%)

The dataset comprises a sample of 4,059 young people (aged 16) selected from 65 difference secondary schools from six inner London Education Authorities.

This is a sub-sample from a much larger study undertaken by Goldstein, H., Rasbash, J., Yang, M., Woodhouse, G., et al. (1993) A multilevel analysis of school examination results, *Oxford Review of Education*, 19, pp. 425-433.

The dataset has been specifically prepared to accompany Rasbash, J. et al. (2005) *A User's Guide to MLwiN 2.0* (Bristol, Centre for Multilevel Modelling).

Simple Linear Regression

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Vrband	Student's score in test of verbal reasoning at age 11, coded into 3 categories (1 = top 25%, 2 = middle 50%, 3 = bottom 25%)

The theory we are exploring is whether the main predictor of 'normexam' is a student's prior performance in a reading test at age 11 (the variable 'standlrt')

The response (outcome or dependent) variable in this case is the student's exam score at age 16 ('normexam') that has been converted into a standardised score (with mean = 0 and standard deviation = 1).

The predictor (independent) variable in this case is the student's prior performance in a reading test at age 11 (standlrt) which is also a standardised score)

Simple Linear Regression

It is generally good practice to standardise your response and predictor variables before including them in the model

It helps in the interpretation of the model.

- For example, if all the predictor variables have been standardised then we know that if they all have the value ‘0’ this would represent the average student.
- As such, the constant in the model comes to represent the mean score on the response variable for the average student in the sample.
- If the response variable has been standardised, we can immediately gain a sense of where the ‘average student’ fits in the distribution of the response variable
  - (i.e. if the constant is negative then we know that they have a mean score below the mean for the sample as a whole, and so on).

Simple Linear Regression

Start by looking at the response variable :

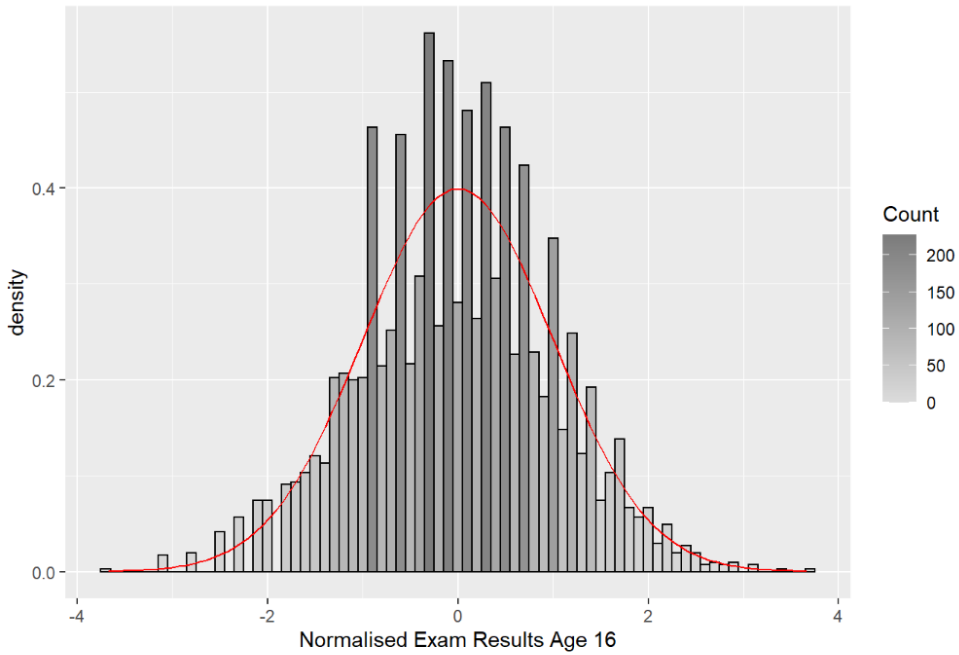
- Create a histogram
- Create a Q-Q Plot
- Examine the skewness and kurtosis and absolute scores
- Create some descriptive statistics

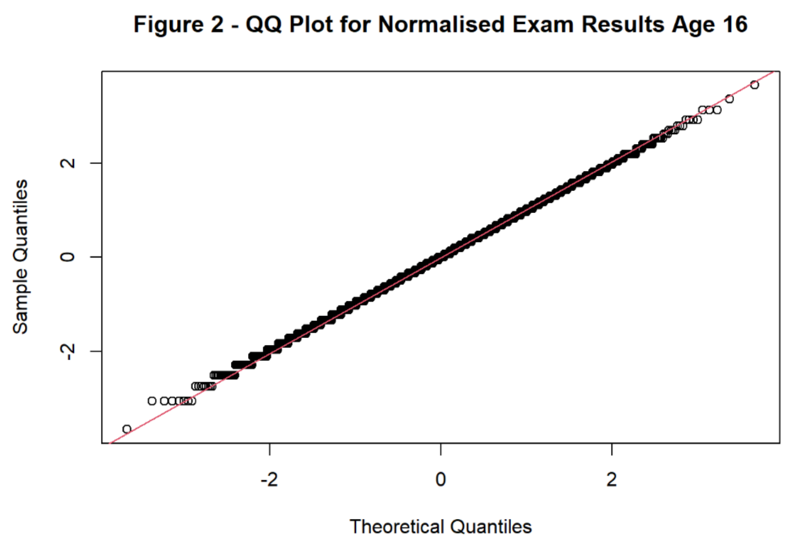
This example:

- The response variable (normexam) is approximately normally distributed
- And approximately standardised (with mean = -0.0001 and standard deviation = 0.9989).

Normexam

Figure 1 - Histogram for Normalised Exam Results Age 16

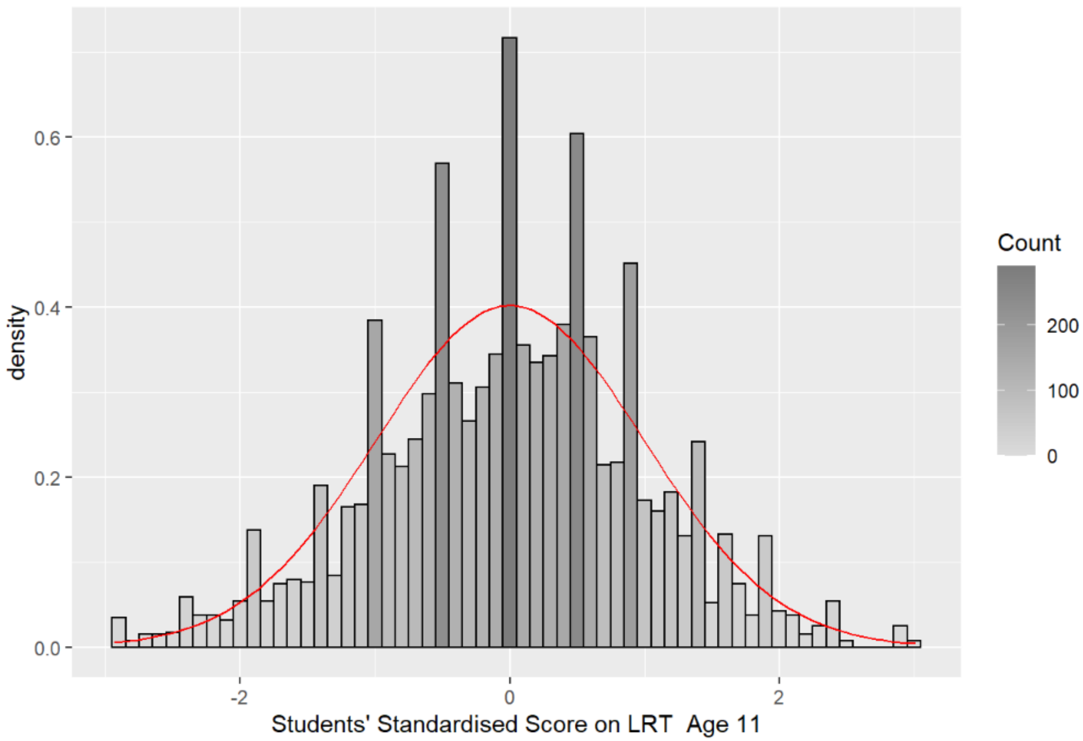


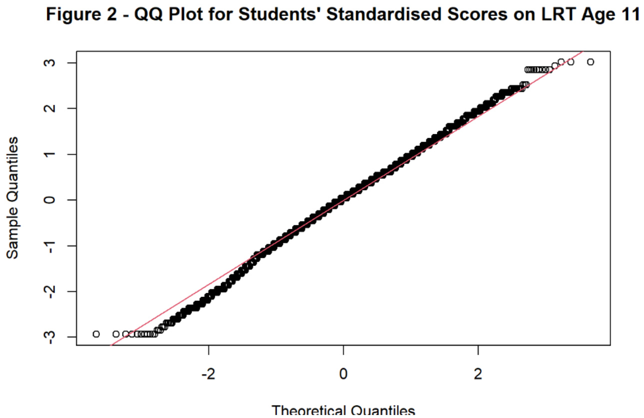


Mean: 0
SD: 1
N: 4059

Standlrt

Figure 1 - Histogram for Students Standardised Score on LRT Age 11



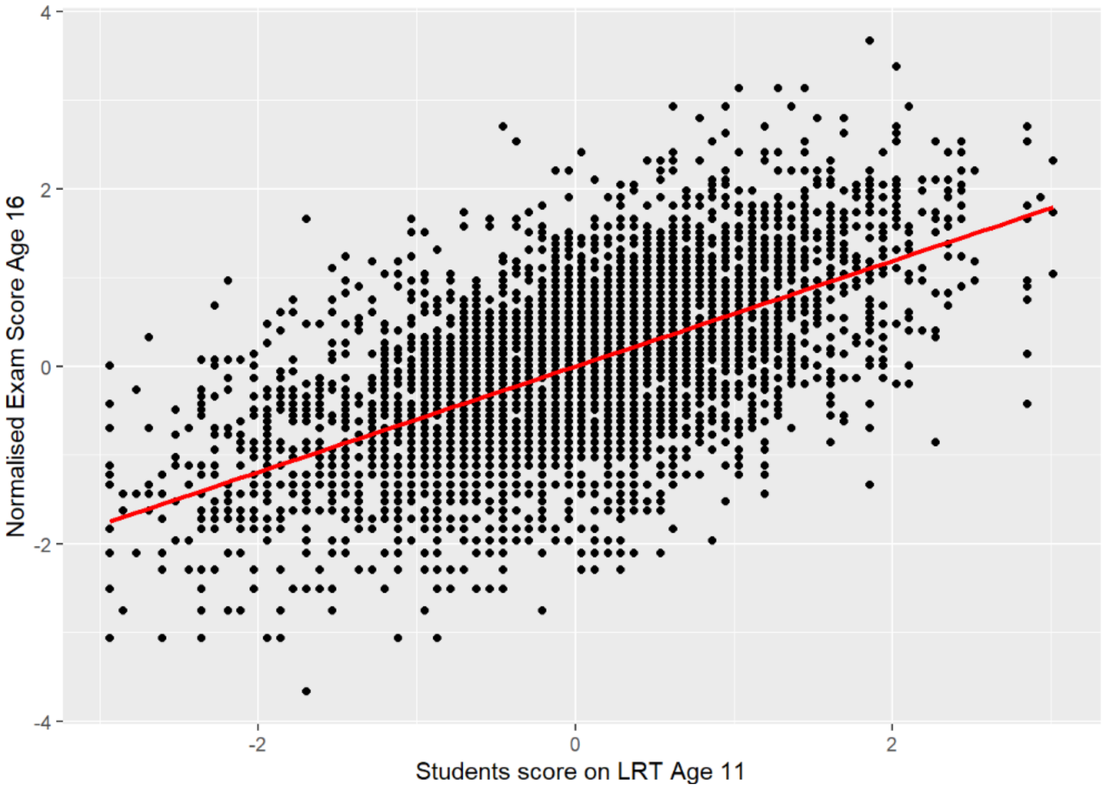


Mean: 0
SD: 1
N: 4059

Simple Linear Regression

Before we create a linear regression model it is worth exploring the nature of the relationship between the response variable and this predictor variable.

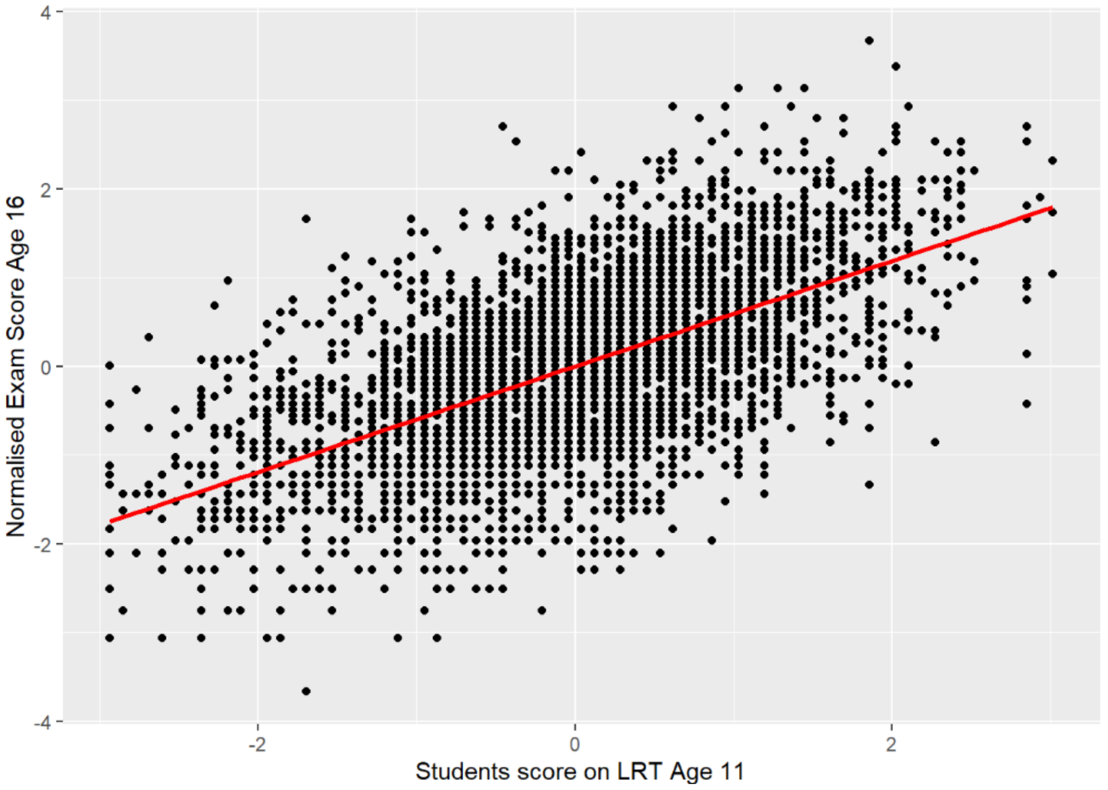
Look at the scatterplot



Simple Linear Regression

Calculate the correlation for these two variables

There appears to be a strong statistically significant positive linear relationship. – R^2 = 0.35, r=0.592, p<0.01




```
	Pearson's product-moment correlation

data:  regression$standlrt and regression$normexam
t = 46.744, df = 4057, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.5712829 0.6112881
sample estimates:
      cor
0.5916496
```

Simple Linear Regression

Suppose we have read our data into a dataframe called regression

To formally model this relationship now we can create a simple linear regression model:

```
#Create the model
model1<-lm(regression$normexam~regression$standlrt)
anova(model1) # Display the output
```

We now have access to all the elements of the regression output in the variable model which we can unpack and examine.

Simple Linear Regression

Increases in scores in standlrt appear to associated with increases in scores in normexam.

There appears to be a moderately strong correlation between the two (r = 0.592, p<0.001).

Therefore, there is good justification for using regression to look at prediction.

The constant represents the value at which the regression line crosses the y-axis.

stargazer(model1, type="text") #Tidy output of all the required stats

```
## ===============================================
##                         Dependent variable:
##                     ---------------------------
##                              normexam
## -----------------------------------------------
## standlrt                     0.595***
##                               (0.013)
##
## Constant                     -0.001
##                               (0.013)
##
## -----------------------------------------------
## Observations                  4,059
## R2                            0.350
## Adjusted R2                   0.350
## Residual Std. Error      0.805 (df = 4057)
## F Statistic         2,185.011*** (df = 1; 4057)
## ===============================================
## Note:               *p<0.1; **p<0.05; ***p<0.01
```

Top value= co-efficient
Lower value = p value

How Good is the Model?

The regression line is only a model based on the data.

This model might not reflect reality.

- We need some way of testing how well the model fits the observed data.

How has the model helped improve our understanding in the absence of any model?

How Good is the Model?

What would we use in the absence of the model?

- The common approach would be to use the mean of the sample
  - This would be wrong in the vast majority of the cases
    - depending on the variation in the sample i.e. how much each case varies from the mean
  - But we can use it as a baseline to see if our model has improved our understanding

How good is the model?

So how do we go about it?

Total sum of squares

- Square the differences from each data point to our mean for our outcome variable
- **total sum of squares** = all variation that exists

Residual sum of squares

- But we will still experience some error
- This is the variation between our observed data and our regression line
- Squaring these errors gives use the **residual sum of squares** (unexplained sum of squares) = variation which is left over after the model is fitted to the data

Difference between the two is the amount of **variation accounted for by the model**

Sums of Squares

SS_T

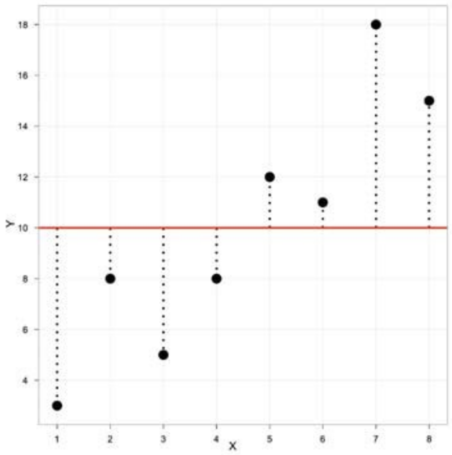
Total variability (variability between scores and the mean).

SS_R

Residual/Error variability (variability between the regression model and the actual data).

SS_M

Model variability (difference in variability between the model and the mean).



Simple Linear Regression

To get this information in R:
`anova(model1)`
`summary(model1)`

```
Analysis of Variance Table

Response: regression$normexam
                      Df Sum Sq Mean Sq F value    Pr(>F)    
regression$standlrt    1 1417.5 1417.50    2185 < 2.2e-16 ***
Residuals           4057 2631.9    0.65                      
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
Residuals:
     Min       1Q   Median       3Q      Max 
-2.65615 -0.51848  0.01264  0.54399  2.97399 

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)    
(Intercept)         -0.001191   0.012642  -0.094    0.925    
regression$standlrt  0.595057   0.012730  46.744   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8054 on 4057 degrees of freedom
Multiple R-squared:   0.35,	Adjusted R-squared:  0.3499 
F-statistic:  2185 on 1 and 4057 DF,  p-value: < 2.2e-16
```

The F statistic looks at whether the model as whole is statistically significant

This allows us to answer the question:

Do the independent variables, taken together, predict the dependent variable better than just predicting the mean for everything?

Simple Linear Regression

SS_M

SS_R

```
Analysis of Variance Table

Response: regression$normexam
                    Df Sum Sq Mean Sq F value    Pr(>F)
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Residuals           4057 2631.9    0.65
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

SS_R

Residual/Error variability (variability between the regression model and the actual data).

SS_M

Model variability (difference in variability between the model and the mean).

SS_T

Total variability (variability between scores and the mean).

Sums of Squares

SS_T

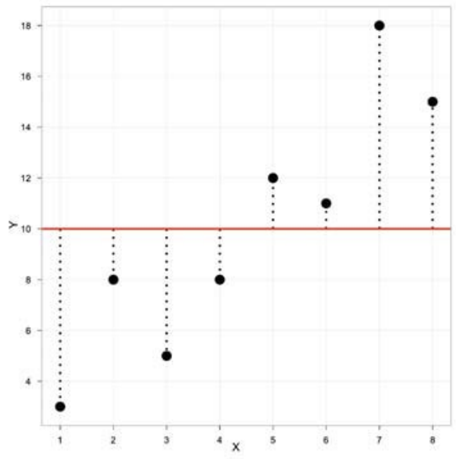
Total variability (variability between scores and the mean).

SS_R

Residual/Error variability (variability between the regression model and the actual data).

SS_M

Model variability (difference in variability between the model and the mean).



Testing the Model: ANOVA

Mean Squared Error

- Sums of Squares are total values.
- They can be expressed as averages.
- These are called Mean Squares, MS

$$\boxed{F = \frac{MS_M}{MS_R}}$$

Simple Linear Regression

MS_M

```
Analysis of Variance Table

Response: regression$normexam
                    Df Sum Sq Mean Sq F value    Pr(>F)
regression$standlrt    1 1417.5 1417.50    2185 < 2.2e-16 ***
Residuals         4057 2631.9    0.65
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

MS_R

MS_M

Mean Square Error of the Model

MS_R

Mean Square Error of the Residuals

Testing the Model: R^2

R^2

- **The proportion of variance accounted for by the regression model**.
- The Pearson Correlation Coefficient Squared

$$\boxed{R^2 = \frac{SS_M}{SS_T}}$$

Simple Linear Regression

anova(model)

```
Analysis of Variance Table

Response: regression$normexam
                     Df Sum Sq Mean Sq F value    Pr(>F)    
regression$standlrt    1 1417.5 1417.50    2185 < 2.2e-16 ***
Residuals           4057 2631.9    0.65                      
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residuals:
     Min       1Q   Median       3Q      Max 
-2.65615 -0.51848  0.01264  0.54399  2.97399 

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)    
(Intercept)        -0.001191   0.012642  -0.094    0.925    
regression$standlrt 0.595057   0.012730  46.744   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8054 on 4057 degrees of freedom
Multiple R-squared:   0.35,	Adjusted R-squared:  0.3499 
F-statistic:  2185 on 1 and 4057 DF,  p-value: < 2.2e-16
```

Simple Linear Regression

```
Residuals:
     Min       1Q   Median       3Q      Max 
-2.65615 -0.51848  0.01264  0.54399  2.97399 

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)    
(Intercept)         -0.001191   0.012642  -0.094    0.925    
regression$standlrt  0.595057   0.012730  46.744   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8054 on 4057 degrees of freedom
Multiple R-squared:   0.35,	Adjusted R-squared:  0.3499 
F-statistic:  2185 on 1 and 4057 DF,  p-value: < 2.2e-16
```

The R^2 is the proportion of variance in the **outcome** (dependent) variable **normexam** which can be explained by the predictor (independent) variables. This is the usefulness of the model.

An R^2 of 1 means the independent variable explains 100% of the variance in the dependent variable. Conversely 0 means it explains none.

Simple Linear Regression

```
Residuals:
     Min       1Q   Median       3Q      Max 
-2.65615 -0.51848  0.01264  0.54399  2.97399 

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)    
(Intercept)         -0.001191   0.012642  -0.094    0.925    
regression$standlrt  0.595057   0.012730  46.744   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8054 on 4057 degrees of freedom
Multiple R-squared:   0.35,	Adjusted R-squared:  0.3499 
F-statistic:  2185 on 1 and 4057 DF,  p-value: < 2.2e-16
```

In this case we have a value of 0.350 which means that standlrt explains 35% of the variance in normexam.

The remaining variance can be explained by variables not currently included in the model.

Simple Linear Regression

```
Residuals:
     Min       1Q   Median       3Q      Max 
-2.65615 -0.51848  0.01264  0.54399  2.97399 

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)    
(Intercept)        -0.001191   0.012642  -0.094    0.925    
regression$standlrt  0.595057   0.012730  46.744   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8054 on 4057 degrees of freedom
Multiple R-squared:   0.35,	Adjusted R-squared:  0.3499 
F-statistic:  2185 on 1 and 4057 DF,  p-value: < 2.2e-16
```

We have only one predictor at the moment so $\mathbf{R^2}$ and adjusted $\mathbf{R^2}$ are the same.

***When we include more predictors, it is the adjusted $\mathbf{R^2}$ we need to report..

Simple Linear Regression

```
Residuals:
     Min       1Q   Median       3Q      Max
-2.65615 -0.51848  0.01264  0.54399  2.97399

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)
(Intercept)         -0.001191   0.012642  -0.094    0.925
regression$standlrt  0.595057   0.012730  46.744   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8054 on 4057 degrees of freedom
Multiple R-squared:   0.35,     Adjusted R-squared:  0.3499
F-statistic:  2185 on 1 and 4057 DF,  p-value: < 2.2e-16
```

Note: The intercept is the constant

The final thing to look at is ‘**Coefficients**’ gives us all the information we need to construct the actual model using unstandardized co-efficients.

The significance levels for each of the main terms (in this case the coefficients associated with the constant and ‘standlrt’) also tell us if there is evidence that each of these terms are adding something to the model (they are statistically significant).

Simple Linear Regression

```
Residuals:
     Min       1Q   Median       3Q      Max 
-2.65615 -0.51848  0.01264  0.54399  2.97399 

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)    
(Intercept)         -0.001191   0.012642  -0.094    0.925    
regression$standlrt  0.595057   0.012730  46.744   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8054 on 4057 degrees of freedom
Multiple R-squared:   0.35,	Adjusted R-squared:  0.3499 
F-statistic:  2185 on 1 and 4057 DF,  p-value: < 2.2e-16
```

Note: The intercept is the constant

Using the Coefficients from this table we get the following model:

Predicted ‘normexam’ = -0.001 + 0.595 x ‘standlrt’

Simple Linear Regression

```
Residuals:
     Min       1Q   Median       3Q      Max 
-2.65615 -0.51848  0.01264  0.54399  2.97399 

Coefficients:
                     Estimate Std. Error t value Pr(>|t|)    
(Intercept)         -0.001191   0.012642  -0.094    0.925    
regression$standlrt  0.595057   0.012730  46.744   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8054 on 4057 degrees of freedom
Multiple R-squared:   0.35,	Adjusted R-squared:  0.3499 
F-statistic:  2185 on 1 and 4057 DF,  p-value: < 2.2e-16
```

What does it mean if the constant (intercept) is negative?

- If constant is negative then when all explanatory variables equal zero, the predicted outcome is below zero. This interpretation is purely mathematical and depends on whether $x_i = 0$ is meaningful in the context.
- If zero is outside the realistic range of the predictors, the intercept itself may not carry practical meaning.
  - The negativity simply ensures that the fitted line or plane best aligns with observed data across the range where predictors are valid.
- It does not indicate a problem with the model unless it makes the predicted values nonsensical.

Simple Linear Regression

```
Call: lm(formula =
regression$normexam ~
regression$standlrt)
Standardized

Coefficients::
(Intercept) regression$standlrt
0.0000000    0.5916496
```

```
Function: lm.beta(model1)
```

Standardised Coefficients

Not hugely useful for our model since the variables are all standardized but will be useful for models you create.

Reporting

A simple linear regression model was created to investigate whether a student's standardized score on the Standard London Reading Test (LRT) at age 11 could be considered to predict a student's standardized exam result at age 16. A statistically significant regression equation was found ($F(1,4057)=2,185.01$, $p < 0.01$) with an adjusted R^2 of 0.35. The regression equation is therefore:

Standardized Exam Score at Age 16 = -0.001 + 0.595 * Standardized score on Standard LRT at age 11

This indicates that for each one-unit increase in standardized LRT score at age 11, the standardized exam score at age 16 is expected to increase by approximately 0.595 units, holding other factors constant. This relationship is statistically significant and suggests a meaningful positive association between early reading performance and later academic achievement.

Time for you to try

Review the section Regression Model 1 in MATH9102-W8-Linear Regression.qmd and the associated html output

Your exercise:

- Use the template MATH9102-W8-RegressionExercise.qmd
- Complete Exercise 1:
  - Using the Sleep dataset from Pallant (sleep.rdata) build a linear regression model to investigate whether:
    - Total Sleep and Associated Sensation score (totsas) can be considered to be predicted by a respondent’s level of anxiety (anxiety) (simple linear regression)
  - You do not need to do the graphs and descriptive statistics – you can assume both can be treated as normal

Multiple Linear Regression

Sources used in creation of this lecture:
Statistics and Data Analysis, Peck, Olsen and Devore; Discovering Statistics Using R, Field, Miles and Field; Understanding Basic Statistics, Brase and Brase;SPSS Survival Manual, Julie Pallant

What is Multiple Regression?

Simple linear Regression is a model to predict the value of one variable from another.

- It is rarely if ever used.

Multiple Regression is a natural extension of this model:

- This is the most common form of regression model used.
- We use it to predict values of an outcome from several predictors.
- It is a hypothetical model of the relationship between several variables.

Multiple Regression as an Equation

With multiple regression the relationship is described using a variation of the equation of a straight line.

$$y = b_0 + b_1 X_1 + b_2 X_2 + \ldots + b_n X_n + \varepsilon_i$$

b0 is the intercept.

- The intercept is the value of the Y variable when all Xs = 0.
- This is the point at which the regression plane crosses the Y-axis (vertical).

b_1 to b_n are the regression coefficient for variable 1 to n

Dummy (indicator) variables

Many variables we are interested in prediction are categorical

- E.g. we are interested in the effect gender or religion, or income category has

Because they have no scale it makes no sense to think of the effect of a unit of increase in these as it would for a continuous variable

But we can think in terms of the **differential effect** for groupings within the category

We can transform the categorical variable into a series of*dummy variables* which indicate whether a particular case has that particular characteristic

- Dummy variables may also be referred to as indicator variables

Our regression dataset (Regression.sav)

Previously looked at standlrt as a predictor of normexam

We might also be interested in gender and whether it has an influence

- There is significant research that gender has an influence in educational achievement.

We might also be interested in exploring the type of school a student attends (either mixed-gender or single-gender schools) as this might also influence a student's examination performance ('normexam').

We are interested in exploring if there is a *differential effect* for students of different genders and for students of different genders attending different types of school.

Dummy Variables

If we just added them into the model as they are what would happen?

- The values of these two variables would be treated as *real numerical values* rather than just arbitrary numbers representing specific categories.

So, we need to transform these into *dummy variables* before we can add them into the regression model.

Dummy variables

0 (*reference category*) and 1 (*category of interest*)

Aim is to explore if there is a *differential effect* for the category of interest when compared to the reference category

Indicator variable

- Switch effect ON (1) or OFF (0)

Before including in the regression model need to first establish if this makes sense to include as a predictor

- Investigate using an independent t-test.

Dummy variables

0 (*reference category*) and 1 (*category of interest*)

- R lm function automatically does this

Our variable for gender is called girl (named for the category of interest)

- Value 1: boy
- Value 2: girl

In our linear model boy will be treated as the reference category, girl as the category of interest

- In our model boy will be given a value of 0, girl a value of 1
- In this model
  - Boy =0, reference category
  - Girl=1, category of interest
  - We are interested in exploring if there is a difference for girls *in comparison to* boys

Adding Girl to the model

```
model2<-
lm(regression$normexam~regr
ession$standlrt+girl)

anova(model2)

summary(model2)
```

```
Analysis of Variance Table

Response: regression$normexam
                      Df  Sum Sq Mean Sq  F value    Pr(>F)
regression$standlrt    1 1417.50 1417.50 2208.010 < 2.2e-16 ***
regression$girl        1   28.06   28.06   43.704 4.317e-11 ***
Residuals           4056 2603.88    0.64
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

```
Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)         -0.10318    0.01990  -5.184 2.28e-07 ***
regression$standlrt  0.59060    0.01268  46.571  < 2e-16 ***
regression$girlgirl  0.16996    0.02571   6.611 4.32e-11 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8012 on 4056 degrees of freedom
Multiple R-squared:  0.357,	Adjusted R-squared:  0.3567
F-statistic:  1126 on 2 and 4056 DF,  p-value: < 2.2e-16
```

Adding Girl to the model

```
model2<-lm(regression$normexam~regression$standlrt+girl)

anova(model2)

summary(model2)
```

```
Analysis of Variance Table

Response: regression$normexam
                     Df  Sum Sq Mean Sq  F value    Pr(>F)    
regression$standlrt   1 1417.50 1417.50 2208.010 < 2.2e-16 ***
regression$girl       1   28.06   28.06   43.704 4.317e-11 ***
Residuals          4056 2603.88    0.64                      
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

This is the impact being a girl has on the predicted normexam

Why is the impact of being a girl being shown? Because we have coded it as 1, it is our category of interest

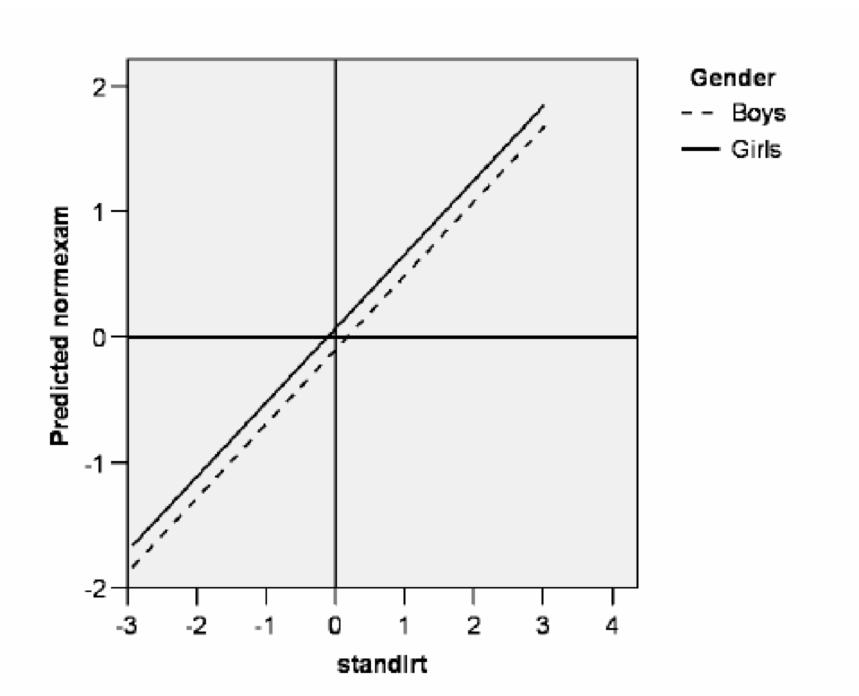
```
Coefficients:
                    Estimate Std. Error t value Pr(>|t|)    
(Intercept)         -0.10318    0.01990  -5.184 2.28e-07 ***
regression$standlrt  0.59060    0.01268  46.571  < 2e-16 ***
regression$girlgirl  0.16996    0.02571   6.611 4.32e-11 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8012 on 4056 degrees of freedom
Multiple R-squared:  0.357,	Adjusted R-squared:  0.3567 
F-statistic:  1126 on 2 and 4056 DF,  p-value: < 2.2e-16
```

Multiple Linear Regression

Our regression equation

What possible values can girl take here?

$$Normexam=-0.10+0.59*standlrt+0.17*girl$$



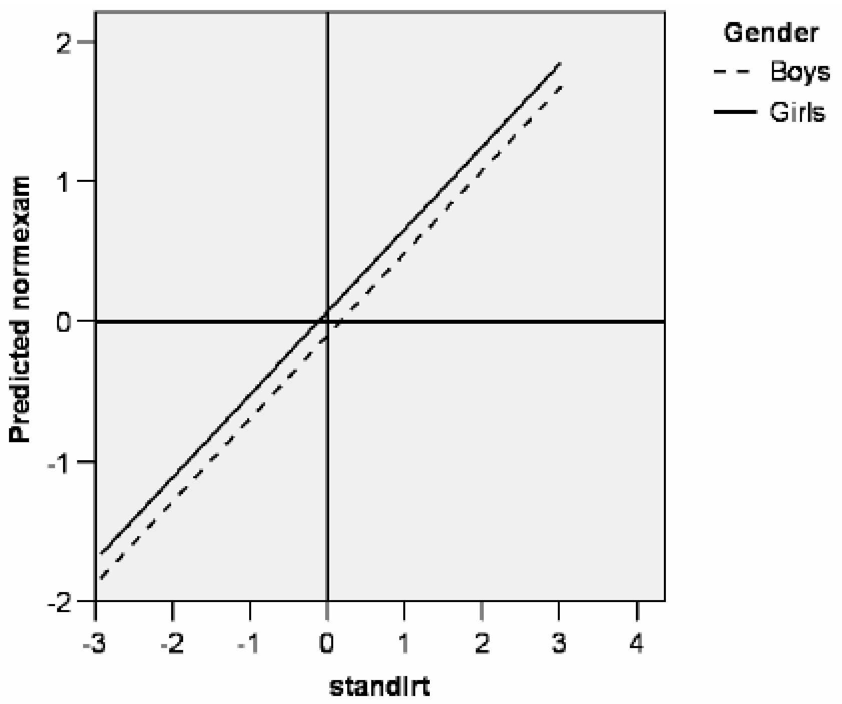
Example

In effect the consequence of adding the dummy variable 'girl' to the model is to create two lines of best fit

- They have the same *gradient* (0.590)
- But different *intercepts* (constants)
  - (i.e. -0.103 for boys (the constant term)
  - and 0.066 for girls (the constant + the co-efficient for girl).

Example

```
Coefficients:
                    Estimate Std. Error t value Pr(>|t|)    
(Intercept)         -0.10318    0.01990  -5.184 2.28e-07 ***
regression$standlrt  0.59060    0.01268  46.571  < 2e-16 ***
regression$girlgirl  0.16996    0.02571   6.611 4.32e-11 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8012 on 4056 degrees of freedom
Multiple R-squared:  0.357,	Adjusted R-squared:  0.3567 
F-statistic:  1126 on 2 and 4056 DF,  p-value: < 2.2e-16
```



- In other words, and as illustrated here, this model can be represented as two parallel lines
- The vertical distance between both lines being 0.170 for the average student (value of 0 for standlrt) when you calculate the equation for boys and girls.

Normexam=-0.10+0.59*standlrt+0.17*girl

Boys normexam = -.103+.590=0.487
Girls normexam=-.103+.590+.167=0.654

Multiple Linear Regression

To investigate the impact gender is having we included our recoded variable into the model

Adding additional predictors has an influence on the other predictors

```
Coefficients:
                    Estimate Std. Error t value Pr(>|t|)    
(Intercept)         -0.10318    0.01990  -5.184 2.28e-07 ***
regression$standlrt  0.59060    0.01268  46.571  < 2e-16 ***
regression$girlgirl  0.16996    0.02571   6.611 4.32e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8012 on 4056 degrees of freedom
Multiple R-squared:  0.357,	Adjusted R-squared:  0.3567 
F-statistic:  1126 on 2 and 4056 DF,  p-value: < 2.2e-16
```

Notice the slight changes in the coefficient for standlrt from the first model

How good a fit is this second model?

In this case our **Adjusted R2** (from the summary) Adjusted R-squared: 0.3567

This model explains 35.67% of the variance in comparison to 35% for model1

Why adjusted R2 ? Because we have multiple predictors

```
Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)         -0.10318    0.01990  -5.184 2.28e-07 ***
regression$standlrt  0.59060    0.01268  46.571  < 2e-16 ***
regression$girlgirl  0.16996    0.02571   6.611 4.32e-11 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.8012 on 4056 degrees of freedom
Multiple R-squared:  0.357,     Adjusted R-squared:  0.3567
F-statistic:  1126 on 2 and 4056 DF,  p-value: < 2.2e-16
```

Multiple Linear Regression

Our regression equation:

Normexam=-0.10+0.59*standlrt+0.17*girl

Lets explore what this means

- For a boy their score for normexam on average increases by 0.49 from their standlrt score:
  - Normexam=-0.10+0.59*standlrt+0 #since girl is 0 for boys
- For a girl their score increases by 0.66 from their standlrt score:
  - Normexam=-0.10+0.59*standlrt+0.17
  - So being a girl is on average adding 0.17 to your score at age 16
  - So, there is a positive differential effect

Linear Regression

Four important statistics

- F statistic
  - Whether the model as a whole predicts the dependent variable
  - Its statistical significance is the significance of the model
- Regression coefficients (Beta values)
  - Measure the strength and direction of relationships between independent variables and the dependent variance
- Significance scores for the regression coefficients
  - Tell us whether the contribution of each variable is statistically significant
- R^2 statistic or Adjusted R^2 Statistic
  - Measures the model's overall predictive power and the extent to which the variables explain the variation found in the dependent variable

How to Interpret Beta Values

Beta values:

- the change in the outcome associated with a unit change in the predictor.

Standardised beta values:

- tell us the same but expressed as standard deviations.

If we have standardised our variables in advance these will be virtually the same.

Constant or intercept (b0)

The constant can be statistically significant when:

- Baseline Differences Exist:
  - There's a substantial baseline level of Y even when all X variables are zero, indicating that the constant is needed to accurately represent the model.
- Model Fit Adjustment:
  - The intercept accounts for any systematic bias not explained by the predictors.
  - A significant intercept suggests there is a systematic part of Y that exists independently of the predictors.

R^2 V Adjusted R^2

Every time you add a predictor to a model, the R-squared increases, even if due to chance alone. It never decreases.

- Therefore, a model with more terms may appear to have a better fit simply because it has more terms.

If a model has too many predictors it begins to model the random noise in the data.

- This condition is known as overfitting the model and it produces misleadingly high R-squared values and a lessened ability to make predictions.

Adjusted R^2

- Adjusts value of R^2 based on number of variables in the model
- Report Adjusted R^2 for multiple linear regression

Reporting

A multiple linear regression model was created to investigate whether a student's standardised score on the LRT at age 11 could be considered to predict a student's standardised exam result at age 16 and whether a differential effect exists for students of different gender. A statistically significant regression equation was found ($F(2,4056)=1,125.86$, $p < 0.01$) with an adjusted R^2 of 0.36:

Standardized score at age 16 = -0.103 + 0.591* Standardized score on Standard LRT + 0.17 * gender (coded as 0 or 1 for girls)

This indicates that an increase of 1 unit in a student's standardized score on the Standard LRT at age 11 can be consider to contribute a statistically significant increase of 0.591 to a students standardized exam result at age 16. A statistically significant differential effect was found for students identifying as a girl which can be considered to contribute an additional statistically significant increase of 0.17 to a student's standardized exam result at age 16.

Reporting

Comparison

A comparison between the simple and multiple linear regression models shows that including gender slightly improves the model's explanatory power. In Model 1, students' standardized scores on the Standard London Reading Test (LRT) at age 11 significantly predicted their standardized exam results at age 16, $F(1, 4057) = 2185.01$, $p < .01$, with an adjusted $R^2 = .35$. In Model 2, when gender (0 = boy, 1 = girl) was added as an additional predictor, the model remained statistically significant, $F(2, 4056) = 1125.86$, $p < .01$, and the adjusted R^2 increased to .36. Both predictors were statistically significant: the standardized LRT score ($\beta = 0.59$, $p < .01$) and gender ($\beta = 0.17$, $p < .01$). This indicates that, controlling for prior reading performance, girls scored on average 0.17 standard deviations higher than boys on their standardized exams at age 16. While the improvement in model fit is modest ($\Delta R^2 = .01$), it suggests that gender contributes a small but meaningful effect on academic outcomes beyond prior reading ability.

Transformed Data

If you conduct your inferential statistics (e.g. linear regression) using transformed data, your interpretation of the equation will change

For linear regression:

- If you use raw values (original unit of measurement) you can say that the co-efficient for each predictor represents the unit of change in the outcome variable for each 1 unit of change in the predictor (when controlling for all other predictors)
- If you transform your outcome variable using sqrt for example and use the transformed values in your equation, you can say that the represents the square root unit of change in the outcome variable for each 1 unit of change in the predictor (when controlling for all other predictors).

Time for you to try

Review the section Regression Model 3 in MATH9102-W8-Regression.qmd and the HTML output

Use the template MATH9102-W8-RegressionExercise.qmd

- Complete Exercise 2:
  - Using the Sleep dataset from Pallant (sleep.sav) you build a linear regression model to investigate whether:
    - Extend your previous model to investigate whether a differential effect exists for respondents who report trouble sleeping v those who do not.
    - Include a dummy variable for trouble sleeping (trubslep)
      - R will dummy code this for you
    - You do not need to do the graphs and descriptive statistics – you can assume both can be treated as normal
    - You do not need to do the difference test at this point.

Including Interaction term

Enable you to examine whether the relationship between the target and the independent variable changes depending on the value of another independent variable.

An interaction term is effectively a multiplication of the two features that you believe have a joint effect on the target.

The following equation represents a model including an interaction term:

$$y = b_0 + b_1 X_1 + b_2 X_2 + \boldsymbol{b_3 X_1 X_2} \ldots + b_n X_n + \varepsilon_i$$

Interaction effect

When the effect of one independent variable on the dependent variable changes depending on the level of another independent variable.

- In other words, it indicates that the impact of one predictor variable on the outcome variable is influenced by the value of another predictor variable.

The presence of an interaction effect suggests that the relationship between an independent variable and the dependent variable is not simply additive.

- Instead, the effect of one variable depends on the level of another variable. (Non-additivity)

Interaction effect

To include interaction effects in a regression model, you typically create a new variable that is the product of the two interacting variables.

- For example, if you have two variables, X1 and X2 , the interaction term would be X1 ×X2
- In R, you can specify an interaction term in a model formula like this:
  - lm(y ~ x1 * x2) # This includes x1, x2, and their interaction

The coefficients of the interaction terms indicate how much the effect of one variable changes with respect to the other variable.

- If the interaction term is significant, it suggests that the main effects of the individual predictors are not sufficient to describe their relationship with the outcome variable.

Interaction term

This model is fine, but it does assume that the gap in exam scores at age 16 between boys and girls remains constant (at 0.170 points) regardless of a student's prior educational achievement at 11 ('standlrt').

But it may be that high achieving boys at age 11 could be outperforming girls at age 16 and/or that low performing boys at age 11 may fall even further behind their female counterparts by the age of 16?

- We are hypothesising here, in effect, that the gradients of both these lines of best fit might not be the same and thus these two lines might not actually be parallel

So, to really improve this we would need to look at including ***an interaction term***

We are investigating here an interaction effect

Interaction term

We can test this hypothesis by adding what is referred to as an '*interaction term'* to the model that is basically a new variable calculated by multiplying 'standlrt' by 'girl'.

- First, therefore, we need to create a new variable (which we can call 'standlrt_girl')

```
regression$interaction <- regression$standlrt *
as.numeric(regression$girl)


*Note: This is already in our dataset regression.sav
```

Model 3 – including an interaction term

	Dependent variable:
	normexam
standlrt	0.593***
	(0.019)
girlgirl	0.170***
	(0.026)
interaction	-0.004
	(0.025)
Constant	-0.103***
	(0.020)
Observations	4,059
R2	0.357
Adjusted R2	0.357
Residual Std. Error	0.801 (df = 4055)
F Statistic	750.399*** (df = 3; 4055)
Note:	*p<0.1; **p<0.05; ***p<0.01

General Equation

Predicted normexam=-.103+0.593*standlrt+0.17*girl-0.0*interaction

Equation for Boys

Predicted Boys normexam=-.103+0.593*standlrt=0.49

Equation for Girls

Girls normexam=-.103+(0.593-0.004) standlrt+.17= -.103+.589*standrt+.17

Interaction term

We can see how the inclusion of the interaction term has basically had the consequence of giving us two lines of best fit (one for boys and one for girls) that now have differing intercepts as well as differing gradients or slopes.

On this occasion we already knew that the difference in the slopes was minor (and not statistically significant)

This is confirmed by the where the difference in the gradients of two lines of best (0.593 compared to 0.589)

- This is marginal and wouldn't even be obvious to the eye if the two lines were plotted on a graph.

We would conclude that this interaction term has no significant effect on the model.

Reporting

A multiple linear regression model was created to investigate whether a student's prior academic achievement, as represented by the standardized score achieved on the LRT at age 11, could be considered to predict a student's standardized exam result at age 16, whether a differential effect exists for students of different genders and whether there is an interaction effect between a student's gender and their prior academic achievement. A statistically significant regression equation was found ($F(3, 4055) = 750.399$, $p < 0.01$) with an adjusted R^2 of 0.357. The regression equation can be expressed as follows:

Standardized exam result at age 16 = − 0.103 + 0.593 × Standardized score on Standard LRT + 0.170 × gender − 0.004 × interaction of prior achievement and gender

This indicates that an increase of 1 unit in a student's standardized score on the Standard LRT at age 11 is associated with a statistically significant increase of 0.593 in the student's standardized exam result at age 16 ($p < 0.01$). Identifying as a girl (coded as 1) contributes an additional statistically significant increase of 0.170 to a student's standardized exam result at age 16 ($p < 0.01$). The interaction term, which represents the combined effect of the standardized score on the LRT and the gender variable, has a coefficient of -0.004, and indicates a non-significant interaction effect ($p = 0.25$). This suggests that the effect of prior academic achievement does not significantly differ between genders in this model.

Reporting - Comparison

This multiple linear regression analysis was conducted to examine whether including an interaction term between gender and prior academic achievement (LRT score at age 11) would improve the prediction of standardized exam results at age 16. An initial regression analysis including standardized LRT score and gender (coded 0 = boy, 1 = girl) as predictors was found to be statistically significant, $F(2, 4056) = 1125.86$, $p < .01$, with an adjusted $R^2 = .357$. Both predictors were significant: higher LRT scores were associated with higher exam results ($b = 0.591$, $p < .01$), and girls scored, on average, 0.17 standard deviations higher than boys ($b = 0.170$, $p < .01$), controlling for prior reading achievement. When an interaction term between gender and LRT score was added, the overall model remained statistically significant, $F(3, 4055) = 750.40$, $p < .01$, but the adjusted R^2 did not change ($R^2 = .357$). The interaction term itself was not statistically significant ($b = -0.004$, $p = .87$), indicating that the relationship between prior reading ability and later exam performance does not differ significantly between boys and girls. In summary, both prior achievement and gender were important predictors, but the inclusion of the interaction term provided no additional explanatory power.

Interaction term

This example has created an interaction term using a categorical variable and a numerical variable

However, you can create interaction terms for two numerical features as well.

Time for you to try

- Review the section Regression Model 3 in MATH9102-W8-Regression.qmd and the HTML output
- Use the template MATH9102-W8-RegressionExercise.qmd
- Complete Exercise 3:
  - Using the Sleep dataset from Pallant (sleep.sav) you build a linear regression model
  - Extend your previous MLR to include an interaction term - number of caffeine drinks per day (caffeine) and the number of alcoholic drinks per day (alchohol – it is misspelt in the dataset).
  - You do not need to do the difference test at this time.

Dummy variables with more than two categories

We need to create several dummy coded variables (the number of which will be one less than the number of categories in the original variable).

We need to define one of the categories as the reference category.

Which category you pick is entirely up to you, but it should make sense in the context of your question

For ordinal variables, it is usually best to select the lowest-ranked category as the reference category.

Dummy variables with more than two categories

Suppose we want to include the type of school a student attended in our regression model.

In our regression model this is:

```
regression$schgend which has 3 values
Mixed (1), Boys only(2), Girls only (3)
```

Dummy Variables

To recode the variable we create two new dummy variables:

```
regression$boy_sch <-
ifelse(regression$schgend=="boysch", 1, 0)
regression$girl_sch <-
ifelse(regression$schgend=="girlsch", 1, 0)
```

Numeric Variable → Output Variable	Old and New Values
schgend → boys_sch	1 → 0 2 → 1 3 → 0
schgend → girls_sch	1 → 0 2 → 0 3 → 1

- This has already been done in the regression dataset we are using.
- It is always worth just checking that you have recoded things properly by comparing the old and new variables via simple crosstabs

```
model3<-
lm(regression$normexam~regression$standlrt+regression$girl+regression$boys_sch
+regression$girls_sch)

Stargazer(model3, type="text")
```

```
===============================================
                         Dependent variable:
                     --------------------------
                              normexam
-----------------------------------------------
standlrt                      0.591***
                              (0.013)

girlgirl                      0.133***
                              (0.034)

boys_sch                      0.183***
                              (0.043)

girls_sch                     0.168***
                              (0.033)

Constant                     -0.161***
                              (0.024)

-----------------------------------------------
Observations                   4,059
R2                             0.364
Adjusted R2                    0.363
Residual Std. Error      0.797 (df = 4054)
F Statistic          580.148*** (df = 4; 4054)
===============================================
Note:               *p<0.1; **p<0.05; ***p<0.01
```

Model 4 – including school gender

Model 4 – including school gender

```
==============================================
                        Dependent variable:
                     ---------------------------
                             normexam
----------------------------------------------
standlrt                     0.591***
                              (0.013)

girlgirl                     0.133***
                              (0.034)

boys_sch                     0.183***
                              (0.043)

girls_sch                    0.168***
                              (0.033)

Constant                    -0.161***
                              (0.024)

----------------------------------------------
Observations                   4,059
R2                             0.364
Adjusted R2                    0.363
Residual Std. Error      0.797 (df = 4054)
F Statistic          580.148*** (df = 4; 4054)
==============================================
Note:                *p<0.1; **p<0.05; ***p<0.01
```

Why isn't mixed school appearing? Because it's the reference category

We are exploring the difference between attending a boys only and attending a mixed gender school and the difference between attending a girls only school and a mixed gender school.

From the model we can also compare a girl attending a girls only to a boy attending a boys only

Interpreting the model

	Constant	Standlrt	Girl	Boy_sch	Girl_sch	
Boy attending Boys only	-.16	.59	0	.18	0	0.61
Boy attending mixed school	-.16	.59	0	0	0	0.43
Girl attending girls only	-.16	.59	.13	0	.17	0.73
Girl attending mixed school	-.16	.59	.13	0	0	0.56

Predicted normexam=-.16+.59*standlrt+0.13*girl+0.18*boy_sch+0.17*girl_sch

So, we can see a difference in scores for boys attending boys only schools and girls attending girls only schools.

Reporting

A multiple linear regression analysis was conducted to examine whether a student's prior academic achievement, as measured by their standardized score on the Standard London Reading Test (LRT) at age 11, could predict standardized exam results at age 16, and to assess whether differential effects exist by gender and school type (single-gender versus mixed-gender schools). The overall model was statistically significant, $F(4, 4054) = 580.15$, $p < .01$, with an adjusted $R^2 = .363$, indicating that approximately 36% of the variance in exam performance was explained by the predictors.

The resulting regression equation was:

Standardized exam result at age 16 = − 0.161 + 0.591 × Standardized score on Standard LRT + 0.133 × gender (girl) + 0.183 × boys school + 0.168 × girls school

This indicates that each one-unit increase in a student's standardized LRT score at age 11 was associated with a statistically significant increase of 0.591 in their standardized exam result at age 16 ($p < .01$). Identifying as a girl (coded 1) contributed an additional 0.133 to the predicted exam score ($p < .01$), relative to boys. Attending a boys-only school was associated with a 0.183 higher exam score ($p < .01$) compared to students in mixed-gender schools, while attending a girls-only school was associated with a 0.168 higher score ($p < .01$). These findings indicate that prior academic achievement, gender, and school type all make statistically significant contributions to predicting later exam performance.

Reporting - Comparison

A multiple linear regression analysis was conducted to examine whether a student's prior academic achievement, as measured by their standardized score on the Standard London Reading Test (LRT) at age 11, could predict standardized exam results at age 16, and to assess whether differential effects exist by gender and school type (single-gender versus mixed-gender schools). The overall model was statistically significant, $F(4, 4054) = 580.15$, $p < .01$, with an adjusted $R^2 = .363$, indicating that approximately 36% of the variance in exam performance was explained by the predictors.

The resulting regression equation was:

Standardized exam result at age 16 = − 0.161 + 0.591 × Standardized score on Standard LRT + 0.133 × gender (girl) + 0.183 × boys school + 0.168 × girls school

This indicates that each one-unit increase in a student's standardized LRT score at age 11 was associated with a statistically significant increase of 0.591 in their standardized exam result at age 16 ($p < .01$). Identifying as a girl (coded 1) contributed an additional 0.133 to the predicted exam score ($p < .01$), relative to boys. Attending a boys-only school was associated with a 0.183 higher exam score ($p < .01$) compared to students in mixed-gender schools, while attending a girls-only school was associated with a 0.168 higher score ($p < .01$). These findings indicate that prior academic achievement, gender, and school type all make statistically significant contributions to predicting later exam performance.

Time for you to try

- Review the section Regression Model 4 in MATH9102-W8-Regression.qmd and the HTML output
- Use the template MATH9102-W8-RegressionExercise.qmd
- Complete Exercise 4:
  - Using the Sleep dataset from Pallant (sleep.sav) you build a linear regression model
  - Extend your previous MLR model which used a dummy variable to represent trouble sleeping to include education level (highest education level completed) as a predictor.
  - In R lm will automatically code edlevel as dummy variables.
    - Primary will be the reference category.